

Data Science

Survival Skills

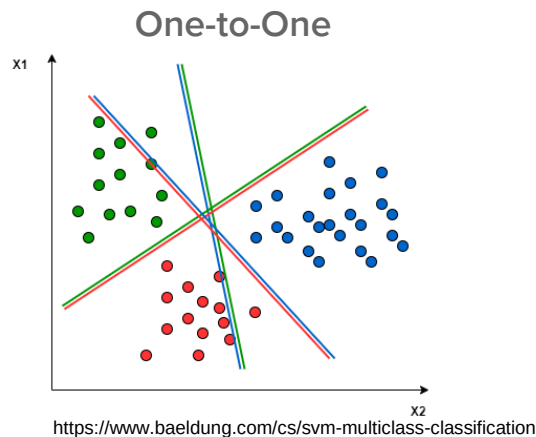
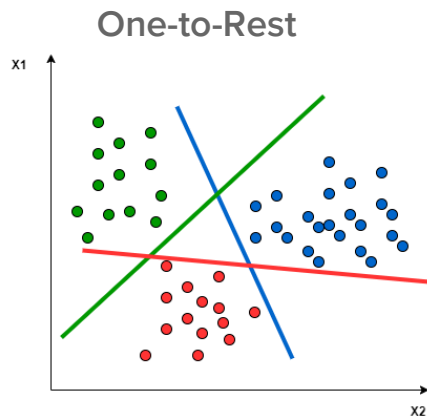
Exercise 8

Questions from last week

- **Bonuspoints:**

0-4 bonus points:	-0.0
5-8 bonus points:	-0.3
9+ bonus point:	-0.7

- **SVM for multiclass problems**



MLP/CNN

Segmentation

Semantic segmentation

**Semantic
Segmentation**



GRASS, CAT,
TREE, SKY

No objects, just pixels

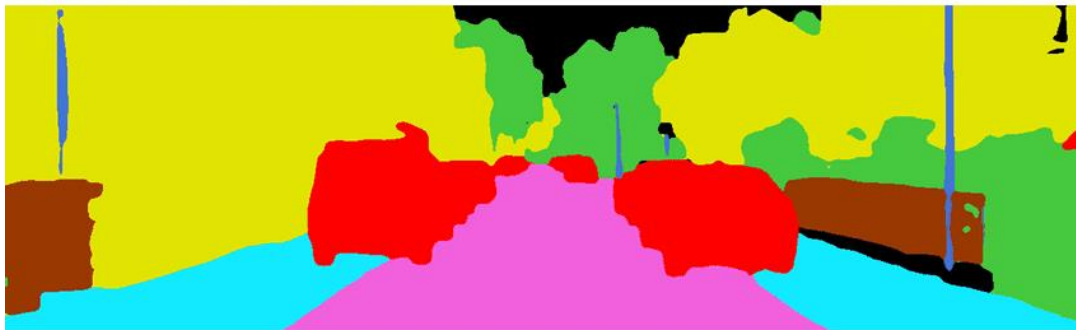
**Classification
+ Localization**



CAT

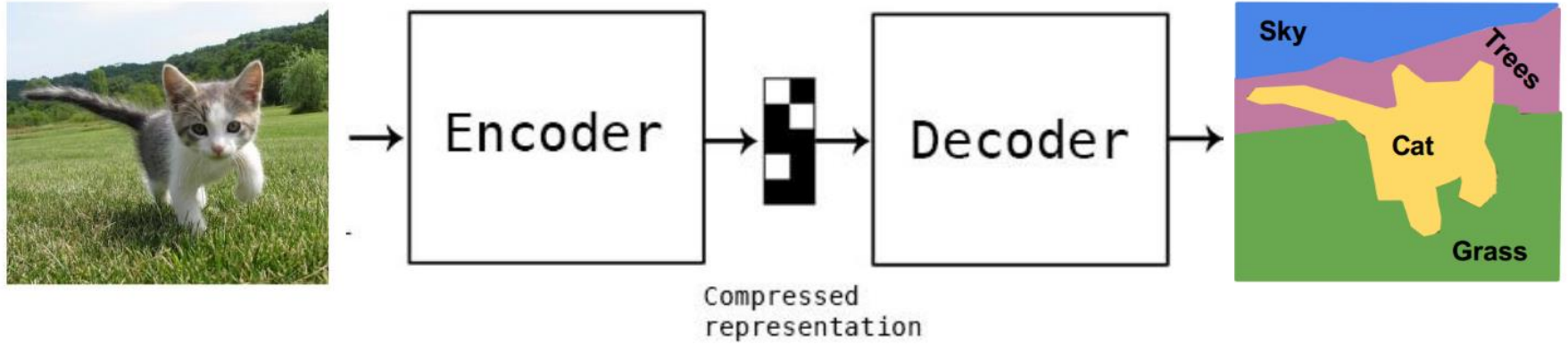
Single Object

Semantic segmentation

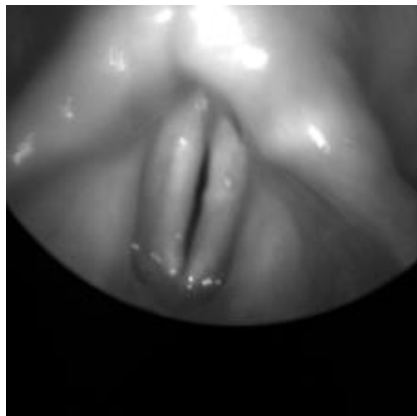


 Road	 Sidewalk	 Building	 Fence
 Pole	 Vegetation	 Vehicle	 Unlabel

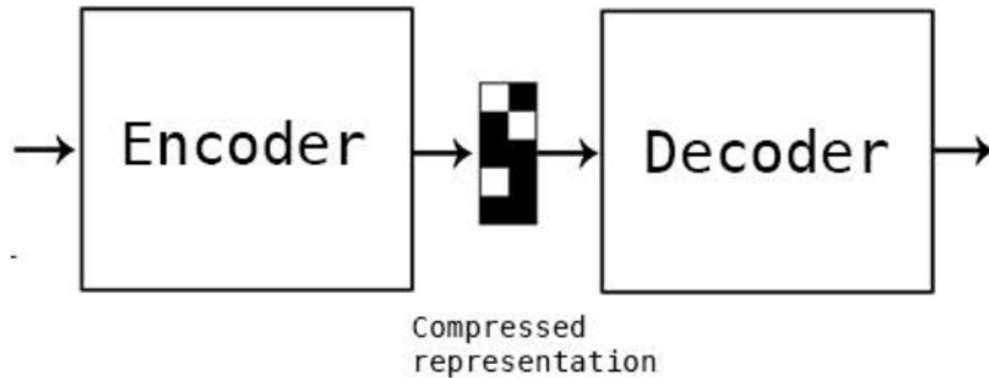
Encoder-decoder network



Encoder-decoder network

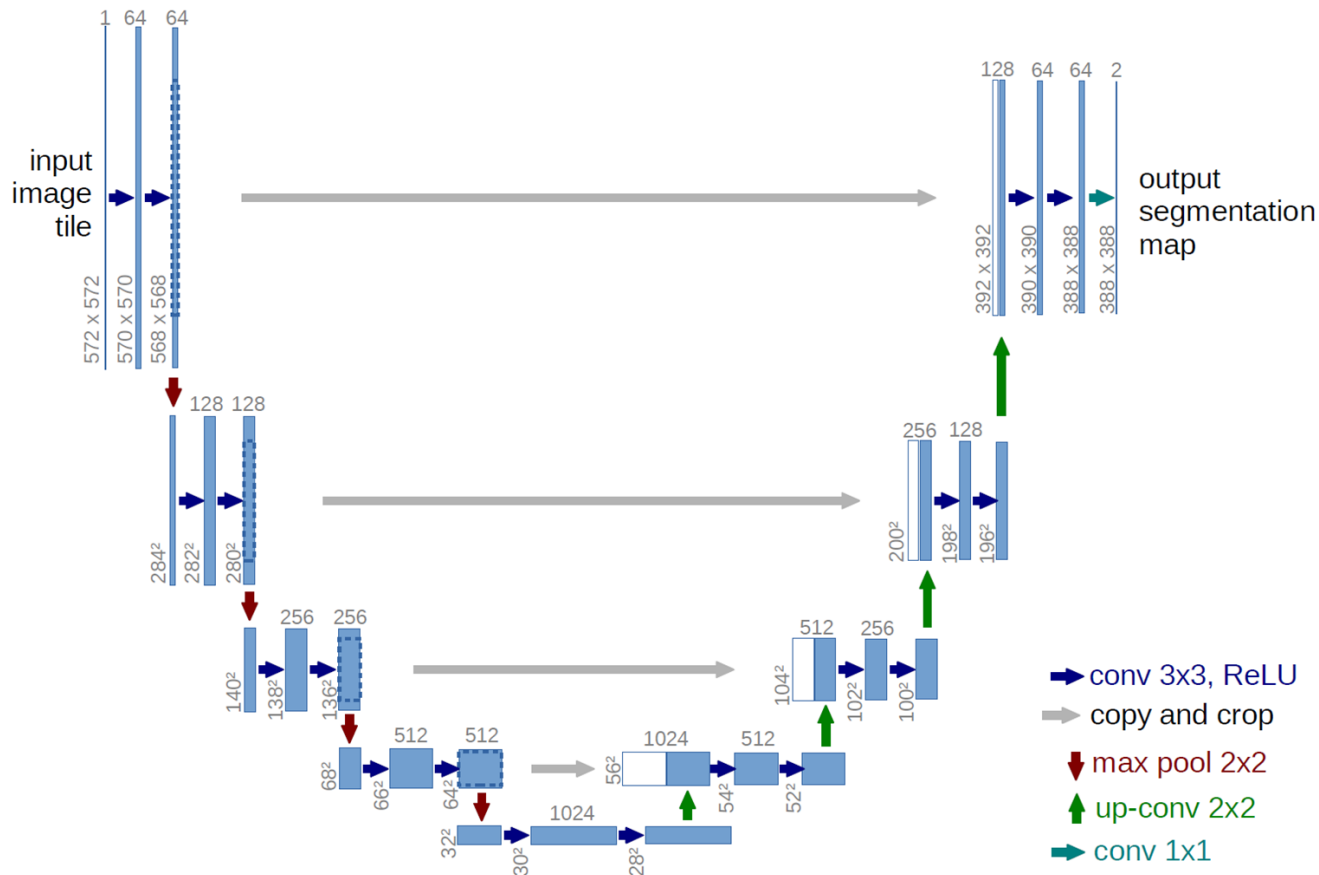


Endoscopic image of
vocal folds & glottis



Glottis segmentation

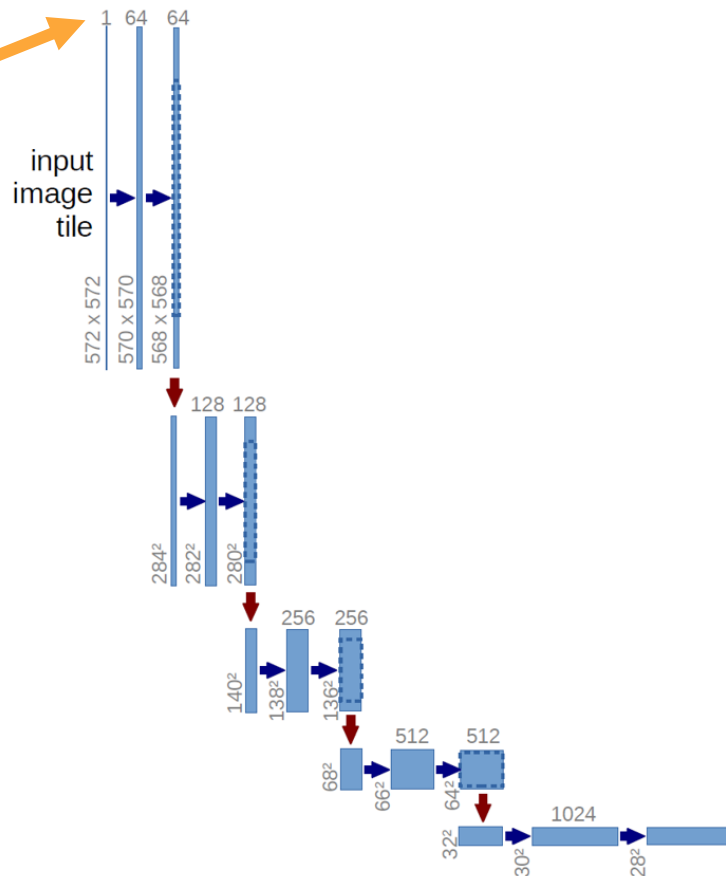
U-Net



U-Net

encoder (downsampling)

channels

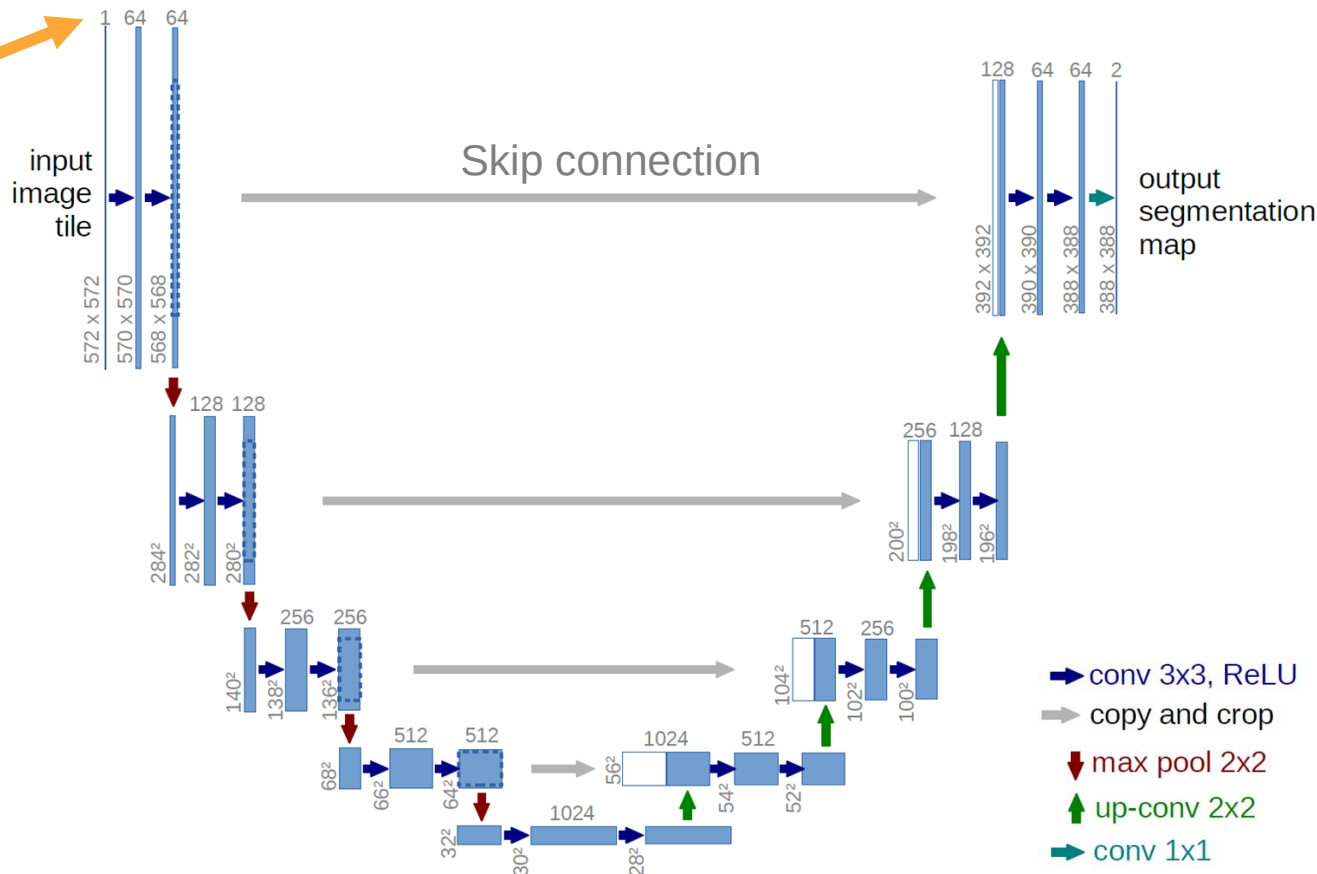


U-Net

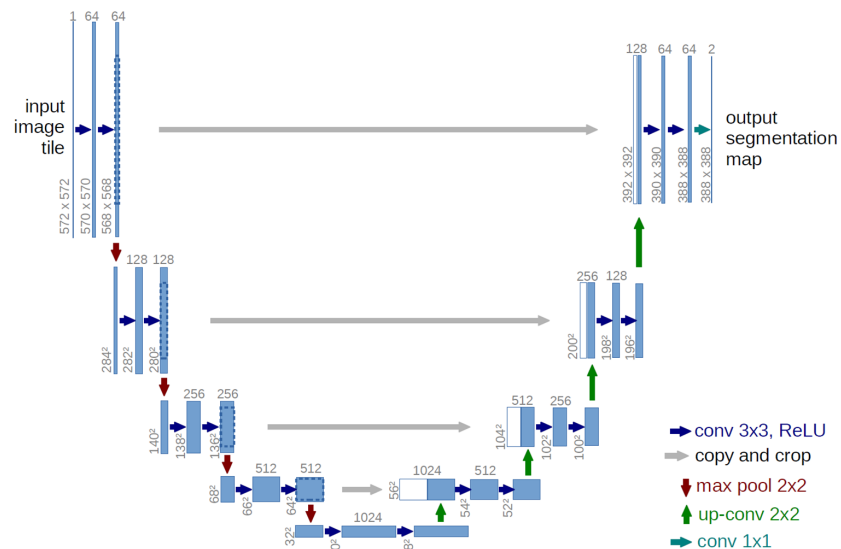
encoder (downsampling)

decoder (upsampling)

channels



U-Net



Reference: Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation." *International Conference on Medical image computing and computer-assisted intervention*. Springer, Cham, 2015.